

THE DIPLOMATIC SIMULATOR

A MULTI-PARTY NEGOTIATION SIMULATOR DRIVEN BY ARTIFICIAL INTELLIGENCE AGENTS

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NYU Center for Global Affairs

July 2026

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Multi-party negotiation is difficult to teach and almost impossible to rehearse. The Diplomatic Simulator asks what happens when the delegations at a crisis table are played by artificial intelligence agents instead of by people. Each delegation is given a confidential brief and nothing else, the talks are run in rounds, and every word spoken is preserved, tagged, and scored. It is a teaching and exploration tool, not a forecasting system, and there is no ground truth anywhere in it.

KEY FIGURES

133

statements of record produced by forty-two delegation agents across six multi-party scenarios

17

negotiating tactics in the fixed vocabulary delegations use to label their own moves

40

Monte Carlo trials run for each scenario under randomly varied shocks, pressure, and mood

0

comprehensive settlements in any trial of any scenario, a ceiling the scenario packs impose rather than a finding about diplomacy

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EXECUTIVE SUMMARY

The Diplomatic Simulator replays multi-party diplomatic negotiations in which every national delegation is played by an artificial intelligence agent, that is, a language program instructed to speak and reason in the voice of one government. It is a teaching and exploration tool. It is not a forecasting system, and its authors state plainly that it must not be used to inform real policy, negotiation, or intelligence judgements.

Six scenarios have been simulated to date: the Arctic, with seven delegations; Central Asia and the Fergana Valley, with seven; Cyprus reunification, with six; the South China Sea, with nine; the Korean Peninsula Six-Party Talks, with six; and Jammu and Kashmir, with seven. An earlier two-party session on the Strait of Hormuz, involving Iran and the United States, is preserved as the original demonstration. Across the six multi-party scenarios, forty-two delegation agents produced one hundred and thirty-three statements of record.

The organising principle of the design is information isolation. When a delegation agent writes its statement for a round, it receives only its own confidential brief, a neutral public brief that everyone shares, and the public transcript of what has already been said aloud. It never sees another government's private instructions, and it never sees another delegation's

private reasoning. This mirrors the informational structure of real diplomacy, where the public record is common property and the mandate is not.

The scenario material is not invented. The confidential briefs are adapted from crisis-negotiation exercise packs associated with the United States Army War College International Strategic Crisis Negotiation Exercise programme, which are taken to universities. The file names preserved in the repository record where each pack was used, including New York University, Penn, Syracuse, and the University of Minnesota.

After the talks conclude, a separate analyst agent reads the whole transcript and produces a scoreboard, a debrief for each delegation, and a neutral summary written in the voice of a convening envoy. A further layer re-runs each scenario forty times under randomly varied external conditions, in order to show which outcomes are stable and which depend on luck.

The repository is candid that every number on the site is one artificial intelligence system's subjective judgement of another artificial intelligence system's writing. There is no ground truth anywhere in the tool. The most serious structural limitation is that a single underlying model plays every side of every table, so the adversaries are not genuinely independent minds. That limitation has now been tested once, on the Korean scenario, by re-running it with different models behind four of the six seats. The coarse outcomes replicated. The test also established something this report had not seen: on that scenario a comprehensive settlement is impossible by construction, because the concessions the parties are authorised to make do not overlap, so the zero-settlement result measures the scenario pack rather than the negotiation. The Limitations section sets out what the comparison does and does not license.

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BACKGROUND AND RATIONALE

The problem. Crisis-negotiation exercises are among the most effective ways to teach diplomacy, because they force a participant to argue a position they may not hold, under time pressure, against people who have been told to resist them. But an exercise consumes an enormous amount of scarce human attention. A seven-party Arctic exercise needs seven teams, seven mentors, and two days. It can be run perhaps once a year at any given institution.

The second problem is that exercises leave almost no durable record. What a delegation said in round two, why it said it, and how that changed the shape of the room afterwards is rarely written down in a form anyone can study later. The learning is real but it is locked inside the experience of having been there.

The gap. Existing computational work on negotiation has largely concentrated on games with fixed rules. The best known example is CICERO, the system developed by Meta's Fundamental AI Research team and reported in Science in 2022, which reached human-level play in the board game Diplomacy by combining a language model with strategic planning. That is a considerable achievement, but a board game has a scoring function and a win condition. Real multi-party talks have neither. A separate strand of work has examined what happens when language models are placed in simulated wargames: Rivera and colleagues, publishing at the ACM Conference on Fairness, Accountability, and Transparency in 2024, found that all five models they tested displayed escalation patterns that were difficult to predict. That finding is a caution rather than an endorsement, and this prototype should be read against it.

The response. The Diplomatic Simulator occupies the space between the two. It takes the scenario packs and confidential briefs that human exercises already use, replaces the human delegations with agents, and keeps the entire record. What the tool produces is not a prediction of what governments would do. It is a legible artefact: a transcript, a set of tagged tactics, and a scoreboard, all of which can be read, disputed, and compared against what a human exercise produced from the same brief.

The design commitment that makes this worth doing is transparency about provenance. Every intermediate product of every run is committed to the repository in a readable form: the extracted party profiles, the public brief, the transcript, the analysis. A reader who doubts a score can open the transcript and check what was actually said.

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OBJECTIVES

The prototype is designed to:

1. Reproduce the structure of a multi-party crisis negotiation, including confidential mandates, public plenary statements, coalition formation, and a convening mediator.

2. Preserve information isolation, so that no delegation gains an advantage the corresponding human team would not have had.
3. Produce a complete and inspectable record of each session, including what each delegation said, which negotiating tactics it used, and how an independent analyst judged the result.
4. Express the outcome in variables simple enough for a non-specialist to read, while keeping the reasoning behind each variable visible.
5. Test how sensitive each outcome is to conditions no negotiator controls, by re-running each scenario under randomly varied external shocks and moods.
6. Document its own method and its own failures, including the specific point at which an orchestration run broke and had to be repeated.

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HOW THE SIMULATOR WORKS

The tool runs each scenario through six stages. The first five use the same reusable toolchain, held in the repository under the folder named `sim`; only the source documents change from scenario to scenario. Three of the six stages are ordinary computer code with no artificial intelligence involved at all.

Stage one, reading the source documents. Each scenario arrives as a set of documents: one public scenario brief describing the crisis, plus one confidential instruction file for each country. These are converted from page images into plain text. No judgement is exercised at this stage.

Stage two, building a party profile. One agent is assigned to each country. It reads that country's confidential instructions and nothing else, and distils them into a structured profile. This is the single most important step in the whole pipeline, because the profile is the entirety of what that delegation will know about itself for the rest of the negotiation.

Stage three, writing the public brief. A separate agent reads the shared scenario document and writes the neutral briefing that every delegation will see. It states the situation, lists the issues formally on the table, sets out the procedure, and fixes the vocabulary of negotiating tactics that delegations will use to label their own moves.

Stage four, the negotiation. The delegations negotiate in plenary rounds. In each round, every delegation writes exactly one statement. It is given its own profile, the public brief, and the running public transcript. It is instructed to stay in role, to ground its claims in its brief, to protect its secret bottom lines, and to label the tactics it has just used. The flagship Arctic scenario runs four rounds: an opening plenary, a positioning round, a bargaining and coalitions round, and a closing plenary. The other three scenarios run three rounds, dropping the separate positioning stage. Seven delegations across four rounds produce twenty-eight statements in the Arctic; nine delegations across three rounds produce twenty-seven in the South China Sea.

Stage five, the analysis. A single analyst agent, cast as a neutral control group, then reads the entire transcript together with every delegation's confidential profile. It is the only agent in the system that sees both sides of the information barrier. It produces the scoreboard, the per-delegation debriefs, and the convener report.

Stage six, assembly and publication. Plain computer code stitches the outputs together, standardises the tactic labels, and renders the interactive pages. No model is involved. This matters for auditing: the numbers a reader sees on the published pages are arithmetic performed on agent outputs, not a second layer of interpretation.

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THE VARIABLES, AND WHY EACH ONE EXISTS

Everything the simulator produces rests on a small number of variables, each of which has a plain meaning. When an agent reads a country's confidential instructions, it fills in eleven fields. Together they constitute the delegation's entire identity.

The eleven fields are as follows.

- Delegation role. One sentence describing who this delegation formally is and how much authority it holds. The Russian profile in the Arctic scenario records a foreign ministry delegation mandated to defend Arctic sovereignty while deferring major deviations to the Foreign Minister. This field exists because a negotiator's freedom to concede is itself a variable, and a delegation that must telephone home behaves differently from one that can sign.

- Fundamental principles. The standing commitments the government would assert regardless of this particular crisis. These give the agent something to argue from when the transcript moves somewhere its specific instructions did not anticipate.
- Desired end state, primary. The delegation's definition of victory.
- Desired end state, alternate. Its definition of an acceptable substitute. Having two rather than one is deliberate: a negotiator with only a maximum position cannot trade.
- Key positions by issue. The government's stated line on each numbered issue in the public brief, stored issue by issue. This is what keeps a delegation consistent across rounds and prevents it from quietly abandoning a position it took an hour earlier.
- Red lines. The commitments the delegation states it will not cross. In negotiation practice a red line is a declared non-negotiable limit whose value depends entirely on whether other parties believe it. The simulator makes this measurable, because the analyst later records how many of a delegation's red lines were crossed by others.
- BATNA. A term of art from negotiation theory, introduced by Roger Fisher and William Ury in *Getting to Yes* in 1981. It stands for Best Alternative to a Negotiated Agreement, and it means the outcome a party falls back on if the talks collapse. It is the benchmark against which every proposal should be judged: a rational party accepts nothing worse than its BATNA. Russia's Arctic BATNA is to proceed unilaterally with resource extraction and sea-route regulation backed by its partnership with China. A delegation with an attractive fallback has little reason to concede, and recording it explicitly is what allows an agent to walk away credibly.
- Concessions willing. What the delegation is authorised to give away, and therefore what can be traded. Without this field a negotiation of principled statements never becomes a negotiation of packages.
- Coalition leanings. Which other delegations this one expects to work with or against. This is what allows blocs to form early rather than emerging only by accident.
- Negotiating style. How the delegation speaks: legalistic, conciliatory, blunt. Style is not decoration. In a transcript that will later be judged, tone is part of what is being judged.
- Private instructions. The secret material the delegation must never state verbatim. This field is the reason information isolation matters. If it were shared, the exercise would collapse into a game of open cards.

Every statement in the transcript is stored with six pieces of information: which delegation spoke, which round it was, what kind of statement it was, the full text, the list of tactics the delegation applied, and the list of other delegations it aligned itself with. The last two are what turn a wall of prose into something that can be counted.

Delegations label their own moves from a fixed list of seventeen terms, set in the public brief. The Arctic list runs:

- anchoring
- counter-anchoring
- conditional-offer
- red-line-signaled
- verification-demand
- deadline-pressure
- coalition-building
- issue-linkage
- appeal-to-law
- appeal-to-precedent
- sovereignty-assertion
- freedom-of-navigation-frame
- side-payment
- delay-tactic
- principled-bargaining-frame
- environmental-frame
- indigenous-inclusion-frame

Several of these are established concepts rather than inventions of the project. Anchoring is the practice of stating an extreme opening figure in order to drag the eventual settlement toward it, and it draws on the anchoring effect documented by Amos Tversky and Daniel Kahneman in *Science* in 1974, who found that an arbitrary starting number measurably shifted people's later estimates even when they were paid to be accurate. Issue-linkage means refusing to settle one question except as part of a package with another. A side-payment is a benefit offered outside the disputed issue to buy agreement on it. Principled-bargaining-frame refers to the approach

set out in *Getting to Yes*, which urges parties to argue from interests and objective criteria rather than from fixed positions.

Because agents do not always use the exact word from the list, a short standardising table in the assembly code folds synonyms into the canonical term. Logrolling, package-linkage, and linkage all become issue-linkage. Batna-signaling and red-line-reaffirmation both become red-line-signaled. Face-saving-formula and consensus-appeal both become principled-bargaining-frame. This is a small piece of housekeeping with a real effect on the counts, and it is worth knowing that it happens.

Each tactic label is stored as a detection record carrying a fixed confidence value of 0.8 and a source marked as self-tagging. That number is not a measurement. It is a constant, recording the fact that the label came from the speaker rather than from an independent observer. A reader should treat the tactic counts as a description of what each delegation said it was doing.

The resulting counts are informative in aggregate. In the Arctic session, coalition-building was tagged twenty-five times and red-line-signaled sixteen, against a single appeal-to-precedent. In the South China Sea session, coalition-building appears twenty-four times and conditional-offer fourteen, with delay-tactic appearing only twice. The shape of a negotiation is legible in these numbers before anyone reads a word of the transcript.

The analyst agent then produces four numbers for each delegation.

- Satisfaction, on a scale from 0 to 1. This expresses how well the analyst judges that a delegation achieved the objectives recorded in its own profile. It is scored against that delegation's stated end states, not against any external standard of a good outcome, which is why a delegation can score well in a session that produced no agreement at all. In the Arctic session, Denmark scored 0.75 and Russia 0.38.
- Agreements won, a simple count of the understandings the delegation secured. Canada won six in the Arctic session, the highest of the seven.
- Red lines crossed, a count of the delegation's declared limits that other parties breached. Russia's three in the Arctic session, against zero for Denmark, is the clearest single indicator of why the two scored so differently.

- Self-rating, a separate figure the delegation assigns to its own performance in its debrief. The United States gave itself 3.3 while the analyst assigned it a satisfaction of 0.6. Keeping the two apart is a useful discipline, since the gap between how a delegation rates itself and how a neutral observer rates it is itself diplomatically interesting.

Each debrief also lists the delegation's goals individually. Every goal carries a priority, either critical or high, a status of achieved, partial, or failed, and a short list of evidence quoting the rounds in which the outcome was settled. This is the part of the output that resists being reduced to a number, and it is the part a reader should check before trusting the number.

A later pass added a second layer to each party profile, describing not only what a delegation wants but how it characteristically bargains. A cultural-background note records the institutional register a delegation tends to negotiate in, whether a high-context consensus style that reads the room before committing or a low-context style that argues directly from the text. A named diplomatic style captures the manner a government is publicly associated with, such as China's assertive wolf-warrior posture, the blunt directness often called cowboy diplomacy in the United States, the reserved consensus-seeking common to Japan, or the flexible hedging Vietnam calls bamboo diplomacy. An alliances field lists the real blocs a country belongs to, from NATO and the Quad to the Shanghai Cooperation Organisation and the China-Russia axis, and a key-figures field names the actual head of government and the foreign and defence ministers in office in the middle of 2026, each with a short note on negotiating style.

These additions are characterisation, not prediction. The named individuals are real office-holders drawn from public reporting, and the notes describe how they are publicly discussed, not any claim about private conduct or how a named person would behave at a real table. Where an office could not be reliably confirmed the field was left empty rather than guessed. The layer sharpens a delegation's voice, and for that reason imports more of the model's biases about real people, which is why it is labelled as sourced characterisation wherever it appears.

READING THE RESULTS

Alongside the scoreboard, the analyst writes a neutral summary in the voice of a Special Representative of the Secretary-General, the title given to a senior envoy appointed by the United Nations Secretary-General to convene and mediate. The report carries a headline, a summary, a list of key outcomes, and a list of unresolved issues. The Arctic report concluded that the conference produced a scaffolding of cooperation tracks but no binding settlement, and observed that what broke through was the low-politics agenda, where trust was cheapest, while everything touching sovereignty, jurisdiction, and membership stalled.

The published site replays each session as a negotiation table, with a scenario picker, the full transcript, and the session insights attached to each statement. A separate dashboard page presents each delegation's private reasoning, its proposals, the tactic labels attached to its statements, and the scoreboard, for any chosen scenario.

A third page streams a completed session statement by statement, at adjustable speed, to convey the pace of a plenary. The page states explicitly that this is a replay of a finished simulation and that nothing is being generated in the reader's browser.

Each scenario also has a standalone written report combining the overview, the convener report, the scoreboard, and the complete transcript, for readers who want the record rather than the interface.

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THE SIX SCENARIOS

The Arctic. Seven delegations, four rounds. The issues are the overlapping continental shelf claims of Russia, Canada, and Denmark over the Lomonosov Ridge, which remain before the United Nations Commission on the Limits of the Continental Shelf; the contested legal status of the Northwest Passage and the Northern Sea Route, which Canada and Russia respectively treat as waters under their own control and which the United States regards as international straits; the equal-access provisions of the 1920 Svalbard Treaty and the fisheries zone Norway maintains around the archipelago; militarisation; resources; environment; governance; and the inclusion of Indigenous peoples. The scenario notes that the United States is not a party

to the United Nations Convention on the Law of the Sea, which shapes what legal arguments it can and cannot make.

Central Asia and the Fergana Valley. Seven delegations. The issues are the enclave-riddled borders inherited from Soviet administrative divisions, the sharing of Syr Darya water between upstream and downstream states, hydropower, and the competing influence of outside powers.

Cyprus. Six delegations. The issues are the choice between a bizonal bicomunal federation and a two-state settlement, territory, property, security guarantees, and the disputed gas fields of the eastern Mediterranean. This is the only scenario whose source pack predates the others, and it is the scenario the simulator finds hardest to settle.

The South China Sea. Nine delegations, the largest table. The issues are the competing maritime claims, the application of the Law of the Sea Convention, freedom of navigation, and the long-running attempt to agree a code of conduct between China and the states of the Association of Southeast Asian Nations.

The Korean Peninsula. Six delegations, three rounds, convened in Beijing under a Chinese chair as a resumption of the Six-Party Talks. The parties are China, North Korea, South Korea, Japan, Russia, and the United States. The issues are guarantees of the non-use of force and whether a Korean War peace treaty falls inside the process; whether a North Korea returned to the Non-Proliferation Treaty in good standing may hold a civilian nuclear programme; diplomatic normalisation and whether a formal end to the war precedes or follows denuclearisation; sanctions relief and the timing of humanitarian assistance; complete verified dismantlement in one comprehensive package against a rewarded step-by-step sequence; and light-water reactors under a revived Korean Energy Development Organisation. The convening report recorded that Beijing reopened three negotiating channels and kept the Six-Party framework alive without reaching a settlement.

Jammu and Kashmir. Seven delegations, three rounds, convened in Geneva under a United Nations Special Representative pursuant to Security Council Resolution 2900, and joined for the first time by a delegation from Jammu and Kashmir itself alongside India, Pakistan, China, Russia, the United Kingdom, and the United States. The talks are organised into three chapters. The territory chapter covers sovereignty over the former princely state, whether the Line of Control becomes an international border, and the Sino-Indian claims to Aksai Chin and the Shaksam Valley. The governance chapter covers the end state for Jammu and Kashmir, the status of Article 370, and

the terms of any plebiscite under the troop-withdrawal conditions of Resolution 47. The human rights chapter covers whether violations are investigated internationally or domestically, the status and repatriation of roughly 950,000 displaced people, and access to the camps and the Kashmir Valley. The convening report recorded that Geneva closed with a single converged instrument, a screened humanitarian relief mechanism for the displaced.

Convening plausibility. A separate question is how plausible it is, in the real world of 2026, that these parties would actually convene and bargain in good faith in the format each scenario stages. This is an editorial judgement about real-world posture, offered as a caveat rather than a result: it is not an output of the simulator and it is not a forecast. A scenario can be a first-rate teaching exercise while describing a table that no government would currently join, and the zero-settlement finding partly measures how far each pack sits from a real, willing negotiation. The ratings below are the author's, on a simple scale of low, moderate, and high.

Central Asia, high. The rare case where good-faith multilateral bargaining actually succeeded: Kyrgyzstan, Tajikistan, and Uzbekistan signed the Treaty of Khujand fixing their border tripoint on 31 March 2025, following the Kyrgyz-Tajik delimitation treaty of the same month. Real negotiation in roughly this format is live and productive.

Cyprus, moderate. The leaders and guarantors do convene under the United Nations, and a fresh informal round was in preparation through 2026, but since the collapse of the Crans-Montana talks in 2017 the sides remain split between a bizonal federation and a two-state settlement, so a good-faith settlement of this scope is not currently in reach.

The Arctic, low. The Arctic Council's political track between Russia and the seven Western states has been suspended since Russia's 2022 invasion of Ukraine, and United States pressure on Denmark over Greenland compounds the freeze. A good-faith seven-party table is not currently thinkable, and China is not a Council member.

The South China Sea, low. China declares the 2016 arbitral award null and void and insists on bilateral rather than multilateral resolution, and the nearest real analogue, the ASEAN-China Code of Conduct, has crawled since 2018 without conclusion.

The Korean Peninsula, low. The Six-Party Talks have been defunct since 2009; North Korea has written its nuclear status into its constitution, adopted a hostile-two-states doctrine renouncing unification, and secured through its 2024 partnership with Russia much of what it once sought at the table.

Jammu and Kashmir, low. India treats the territory as strictly internal, a stance hardened after the 2019 revocation of Article 370, and refuses third-party mediation, so a Security Council conference seating a separate Jammu and Kashmir delegation is unthinkable to New Delhi even as Pakistan seeks exactly that.

Four of the six scenarios rate low, which is a caution against reading any negotiation result here as commentary on how these disputes are likely to move. The packs were built to be teachable, which often means describing a table more balanced and more willing than the real one.

08

TESTING UNDER UNCERTAINTY

A Monte Carlo simulation estimates the range of possible outcomes by running a model many times with randomly drawn inputs and counting how often each result occurs. The technique was devised at Los Alamos in 1946 by Stanislaw Ulam and developed with John von Neumann and Nicholas Metropolis, who supplied the name; Metropolis later recorded that it referred to an uncle of Ulam's who kept borrowing money because he just had to go to Monte Carlo.

A single run tells one story. Whether that story was inevitable or a fluke is a different question, and it is usually the more useful one. Saying that a deal fails in two runs out of eight, and specifically when a hardline shock coincides with a confrontational mood, tells a reader far more about fragility than any single result can.

Each trial draws a fresh value for four things no delegation controls.

- External shock, drawn from a list written for each scenario. The Arctic list includes a record ice-free summer opening new routes, a Russian naval incident involving a NATO vessel, and a major hydrocarbon discovery in contested waters. The Central Asian list includes a severe drought collapsing Syr Darya flows and a deadly border clash. The South China Sea list includes a collision at Second Thomas Shoal and the declaration of an air defence identification zone. One option on every list is that no major shock occurs, so that the absence of a crisis is itself a sampled condition.
- Mediator pressure, set to low, medium, or high. This represents how hard the convening envoy pushes for a result.
- Overall mood, set to cooperative, neutral, or confrontational. This represents the atmosphere in the room, which experienced negotiators treat as a real variable rather than a soft one.
- Momentum, recording which delegation enters with the initiative.

An agent simulates the outcome under that specific draw and reports the type of deal reached, each party's satisfaction, whether each party's primary goal was achieved, partial, or failed, whether its red line held, and which coalitions formed. Deal types are recorded on a five-point ordered scale: comprehensive, framework, partial, stalemate, breakdown. Ordinary code, with no model involved, then counts the deal types, computes each delegation's satisfaction as a mean, a standard deviation, a median, and a lowest and highest value, calculates the percentage of trials in which each red line held, and ranks the coalitions by how often they recurred. A wide spread means the outcome depends on conditions; a narrow one means it is robust.

Forty trials were run for each scenario, and twenty for the earlier Strait of Hormuz session. Across every trial of every scenario, not one produced a comprehensive settlement. Central Asia was the most stable, returning a framework agreement in twenty-six trials of forty and a partial in thirteen, with a single stalemate and no breakdowns. The Arctic split evenly, sixteen frameworks and sixteen partials, with seven stalemates and a single breakdown. Cyprus was among the most fragile of the original four, returning twelve stalemates and seven breakdowns against twelve frameworks. Of the two scenarios added later, the Korean Peninsula proved the most fragile table in the whole set, reaching a framework in only four trials against eleven partials, sixteen stalemates, and nine breakdowns, while Jammu and Kashmir returned six frameworks, seventeen partials, ten stalemates, and seven breakdowns. Individual variables are equally revealing: in the Arctic trials, China's red line held in eighteen runs of forty and the United

States red line in thirty-five of forty, while the five other delegations held theirs in all forty. Counts are given rather than percentages throughout, because a percentage taken over forty trials invites a reader to treat it as a rate estimated from a sample, and it is a census of forty runs of one scenario.

The absence of a comprehensive settlement across every trial reads as a finding about diplomacy and is not one, at least on the scenario where it has been examined. The mixed-model comparison described under Limitations checked the Korean profiles against each other and found that the maximum concession the DPRK is authorised to make and the minimum the United States, the Republic of Korea and Japan are authorised to accept do not overlap at any point. The DPRK's red line forbids reducing its deterrent without acceptable compensation; the other three forbid any residual programme whatsoever. On that table a comprehensive settlement is unreachable by construction, so its absence measures the scenario pack rather than the negotiation, and the figure cannot discriminate between two runs, two models, or two sets of tactics. The same check has not been run on the other five scenarios, and until it has, the zero-settlement result should be read as a property the scenarios may impose rather than an outcome the talks produced. What the trials do measure, and measure usefully, is the distribution below that ceiling: how often a table reaches a framework rather than a partial, how often it breaks down, and which red lines survive varied conditions.

Three further variables were added to each trial. The first records the blocs that formed, distinguishing a standing alliance, such as NATO or the China-Russia axis, from a situational partner of convenience, and rating each bloc's cohesion as tight, loose, or fractured, so that a coalition's strength and durability are legible and not only its membership. The second records the form of any agreement, on a scale from none through a communiqué, a memorandum, and a framework to a full treaty, separating the substance of a deal from its legal weight. The third records, for each party, whether it could realistically ratify and deliver the outcome at home, since a deal at the table is not a deal until it survives domestic ratification.

Because a breakdown can be a win for a party that prefers no agreement, the probability of success is reported not as one number but as several: the share of trials reaching any agreement, the share reaching a framework or better, each party's rate of attaining its primary goal, and the rate at which red lines and ratification hold. Across the six scenarios these figures track the fragility ordering seen in the deal types. Central Asia

reaches some agreement in almost every trial and a framework or better in about two thirds of them, while the Korean Peninsula reaches any agreement in fewer than two trials in five and a framework or better in only one in ten. A high in-simulation success rate on a low-plausibility table still describes the scenario pack more than it describes the world.

The methodology page is explicit that each trial is a reduced-form outcome simulation conditioned on the random draw, not a complete re-run of the multi-round negotiation. What the exercise samples is the model's own distribution over plausible outcomes. It is not an empirical distribution of real events, and the randomised conditions are illustrative rather than calibrated probabilities. The right reading is how sensitive the model thinks this outcome is to shocks and mood, and nothing stronger.

09

GROUNDING IN NEGOTIATION THEORY

The variables are not arbitrary. Recording a BATNA, red lines, and a set of tradeable concessions for every party reproduces the standard analytic apparatus of the Harvard Negotiation Project, and it is what allows the simulator to distinguish a delegation that conceded from a delegation that had nothing to gain by holding out.

The distinction between claiming value and creating value, developed in the study of labour negotiations by Richard Walton and Robert McKersie in 1965, appears in the structure of the tactic vocabulary. Anchoring, counter-anchoring, and deadline-pressure are moves that divide a fixed quantity. Issue-linkage, side-payments, and conditional offers are moves that enlarge what is available to divide. The counts therefore say something about what kind of negotiation took place, not merely how much of it there was.

One well-established feature of real diplomacy is deliberately absent. Robert Putnam's account of two-level games, published in *International Organization* in 1988, describes negotiators bargaining simultaneously at an international table and a domestic one, where a leader whose parliament will reject a deal has less room to agree but more leverage to refuse. The simulator captures a trace of this in the delegation role field, which records how much authority a delegation holds, but it does not model the domestic table. No agent faces a legislature, a coalition partner, or an election. This is a substantial simplification of what constrains real negotiators.

The convening role is modelled at a similarly light touch. The public brief provides that a Special Representative chairs the plenary and may invite proposals and summarise, and the analyst writes in that voice afterwards. But no agent plays the mediator during the talks. Mediator pressure appears only as a randomised condition in the Monte Carlo layer, never as an actor making choices in the room.

10

PROVENANCE OF THE SCENARIO MATERIAL

The scenario packs are adapted from crisis-negotiation exercise materials associated with the International Strategic Crisis Negotiation Exercise programme run by the Center for Strategic Leadership at the United States Army War College, which takes the exercise to universities each year and has developed regional scenario sets including the Arctic, Cyprus, and the South China Sea.

The repository preserves the source documents themselves, organised by scenario, with one public scenario brief and one confidential instruction file per country. Their file names record the institution and academic year of the exercise from which each pack came, which is an unusually clean audit trail for material of this kind.

The material is notional exercise content written for teaching. It does not represent the actual negotiating position of any government, and the simulator inherits that status. Search-engine indexing of the published site is disabled, both through a site-wide instruction file and through per-page tags, so that AI-generated statements attributed to real states do not circulate as though they were reporting.

11

LIMITATIONS AND CAVEATS

The repository documents its own limitations at length, and they are serious enough to state in full.

One mind plays every side. All delegations are driven by the same underlying model. Information is isolated, but the adversaries are not genuinely independent actors, and shared assumptions or style can leak across roles.

This is the deepest limitation in the design, because it undercuts the premise that a negotiation is a meeting of different minds.

That confound has now been tested, on one scenario. An independent peer review put it plainly: the monoculture is acknowledged but never measured, and the test is cheap. It has since been run for the Korean Peninsula, as a controlled comparison against the single-model baseline holding scenario, profiles, brief, prompts, rounds and information isolation constant and varying only which model sits behind each seat. The two delegations defending the hardest positions stayed on the strongest model, so that a weaker model self-breaching a red line could not be confused with the position failing; the chair and Russia moved to a mid-tier model and the two allied delegations to the smallest.

What that test found is mostly adverse to this report. The coarse outcomes replicated: substituting four of six seats across two model tiers did not change whether the mutual freeze was accepted, whether sanctions moved, or whether a settlement was reached. Three finer outcomes did not replicate, and the bilateral channel on the abductions agreed in the baseline was never convened in the second run. But the central result is that the zero-settlements figure is over-determined by the scenario pack rather than produced by the talks, and it is set out under the Monte Carlo results above. Two further cautions belong with any citation of the comparison. One headline outcome moved between the two runs at a seat whose model never changed, which is ordinary run-to-run variance rather than an effect of substitution, and it bounds how much weight any single reported difference can carry. And the baseline is not a clean reference: it contains the same families of error as the mixed run, including two delegations asserting a five-party coalition the public record does not support and attributing it to the record.

Model and role cannot be separated in that test. Both seats on the smallest model were the two allied non-American delegations and both seats on the mid-tier model were the two step-by-step delegations, so the model and the chair it sits in are perfectly confounded. The claim that a weaker model degrades role fidelity and the claim that the Seoul and Tokyo seats are simply harder to play are indistinguishable on this data. There is one run in each arm, so nothing in the comparison is a rate, a frequency, or an effect size, and it should not be cited as one. Rotating each model across each seat, replicating enough to turn instances into rates, and re-auditing the transcripts with a coder blind to the model assignment are the work that would license more.

There is no ground truth. Nothing in the tool has been validated against a real negotiation, an expert assessment, or a historical outcome. The scoreboard numbers are one system's subjective judgement of another system's writing. Satisfaction should be read as how the analyst rated the argument on the page, not as an assessment of any national interest.

Agents can fabricate. A language model can invent a plausible-sounding treaty, figure, or precedent that appears nowhere in its brief, and can drift from a government's real position in ways that read smoothly.

Training data shapes the outcome. Which arguments sound strong, which framings feel reasonable, and how sympathetically each actor is portrayed all reflect biases in the material the model learned from. A simulator in which one delegation consistently argues more persuasively than another may be reporting a fact about the model rather than about the dispute.

The exercise is compressed. A multi-day negotiation becomes three or four short plenary rounds. Caucuses, bilateral side-channels, corridor conversations, and escalation dynamics are simplified or absent. Much of what determines the outcome of real talks happens outside the plenary, and none of that is here.

The samples are small. Forty trials per scenario is enough to show that Cyprus is fragile and Central Asia is not, and that the two scenarios added later, Korea and Jammu and Kashmir, are more fragile still. It is not enough to support a percentage figure quoted on its own.

The trial count. The Monte Carlo layer now holds forty trials for each of the six scenarios and a further twenty for the earlier Strait of Hormuz session, for two hundred and sixty in total. Anyone reconciling this figure against an earlier count should note that the run was topped up from an initial eight, then twenty, trials per scenario.

The project records a failure honestly. The first orchestration run struck a transient network outage that stopped roughly half its agents mid-negotiation, and a targeted re-run completed the missing rounds. The repository states that no gaps were papered over and that every published statement is a genuine agent output, with the intermediate artefacts committed for inspection. Because language models are not deterministic, the exact wording of any repeat run will differ even from identical source documents.

The isolation has been audited on every scenario. A cross-party contamination audit now exists for all six multi-party scenarios, tracing each statement to the actor's own brief, the shared public brief, or legitimate public inference, and checking whether any delegation's concealed cards ever surfaced through a rival's mouth. None did: in every scenario the tell-tale secrets stayed dark, and the audits rate information integrity as high. What they flag instead is stylistic homogenisation, the single-author fingerprint of one model writing every side, which is the artefact the mixed-model test is built to probe.

The named figures are a characterisation. Attaching the real head of government and the foreign and defence ministers to each delegation sharpens its voice but also imports the model's biases about specific, living individuals. These entries are drawn from public reporting and describe how each figure is publicly discussed; they are not claims about private conduct, and the simulator does not model any named person's actual decisions.

It is not decision support. The repository states directly that the tool must not be used to inform real policy, negotiation, or intelligence judgements.

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INTENDED AUDIENCE AND USE

The prototype is built for teaching and for structured exploration. Its most defensible use is as a companion to a human exercise rather than a replacement for one. A class that has run the Arctic scenario itself can compare its own transcript against the simulated one and argue about where the agents were wrong, which is a more demanding exercise than either activity alone.

It is also of interest to researchers examining what happens when language models are asked to hold a position under pressure. The run artefacts are complete enough to support that kind of scrutiny, which is precisely what makes the honest limitations section valuable rather than decorative.

It is of no use to anyone seeking to know what any government will actually do. The tool says so itself, and this report repeats it because the output is fluent enough to be mistaken for analysis.

CONCLUSION

The Diplomatic Simulator is a modest and carefully documented prototype. Its contribution is less the simulated negotiations themselves than the discipline imposed on them: confidential mandates that stay confidential, statements that are counted and labelled, scores that are defined against each delegation's own stated goals rather than an invented standard of success, and a sensitivity layer that reports how much of the outcome was luck.

The most valuable thing in the repository may be its methodology page, which sets out what the agents do, what they do not do, which model produced each layer, where the run broke, and why none of the numbers carry ground truth. Tools of this kind will become common, and their outputs will be fluent enough that readers extend them more credibility than they have earned. A prototype that publishes its own failure modes alongside its results, and that turns off search indexing so its invented statements are not mistaken for reporting, sets a standard worth borrowing.

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